Parametric Physics-Constrained Neural Network for 1D Casson Fluid Flow in a Cylindrical Pipe

Omphile N. Seabe

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Abstract

Predicting the flow behaviour of non-Newtonian fluids is a cornerstone of modern engineering and simulation. Traditional Computational Fluid Dynamics (CFD) methods are computationally expensive, while standard data-driven Machine Learning requires vast amounts of experimental data. This project explores a Physics-Informed Neural Network (PINN) approach to bridge this gap, specifically targeting the flow of a Casson fluid in a horizontal pipe. We propose a Parametric PINN architecture that moves beyond single-state simulations. The Casson constitutive relationship and shear-stress formulation derived from the steady-state Navier–Stokes equations were embedded into the neural network's loss function. The network is designed to be parametric, accepting both the radial coordinate (r) and the plastic viscosity (μ_c) as inputs. The performance of the model is also compared with analytical and numerical solutions. Simulations were conducted using Python. The performance of the model was validated through a triple-comparison with analytical solutions and numerical finite-difference methods. Results demonstrate high predictive accuracy, with mean absolute errors in the order of 10^{-3} m/s and precise capture of the characteristic *plug flow* region. Furthermore, the model exhibited robust zero-shot generalization, successfully predicting velocity profiles for unseen viscosity values within the parametric range.

Keywords: Physics-Informed Neural Networks (PINNs), Casson Fluid, Non-Newtonian Flow, Parametric Modeling, Python, PyTorch, Digital Twins.

1 Introduction

The simulation and real-time monitoring of complex fluid dynamics remain a central challenge in modern industrial engineering and "Smart Manufacturing." Traditional computational approaches, such as Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA), rely on the numerical discretization of governing partial differential equations (PDEs). While highly accurate, these methods are often computationally expensive and unsuitable for real-time applications or iterative optimization cycles.

Conversely, purely data-driven machine learning models have emerged as efficient surrogates; however, they require vast amounts of high-fidelity labeled data, which is often unavailable or difficult to acquire in "black-box" industrial environments where physical sensors are restricted.

To bridge this gap, **Raissi et al. (2019)** introduced the framework of **Physics-Informed Neural Networks (PINNs)**. By embedding the governing physical laws—expressed as nonlinear PDEs—directly into the loss function of a deep neural network, PINNs act as differentiable solvers that do not require external labeled data. This methodology leverages the universal approximation power of neural networks while ensuring that the resulting predictions remain strictly consistent with the underlying conservation laws of mass, momentum, and energy.

This project demonstrates the potential of a **Parametric PINN** to act as a **Digital Twin** for non-Newtonian fluid systems. By training a single model to generalize across a continuous manifold of physical parameters (such as viscosity and yield stress), we move beyond static

simulations toward dynamic, real-time surrogates. Such models provide physics-consistent flow predictions that are critical for optimizing energy consumption, ensuring product consistency, and enabling predictive maintenance in complex processing lines where traditional sensing is unfeasible.

2 Mathematical Framework

2.1 Governing Equations and Assumptions

The fluid dynamics of Casson fluid flow in a horizontal pipe are governed by the incompressible Navier-Stokes equations. To arrive at a tractable 1D model suitable for PINN implementation, we assume a steady-state ($\partial/\partial t = 0$), fully developed, and axisymmetric flow ($\partial/\partial\theta = 0$). Under these conditions, the momentum balance in cylindrical coordinates simplifies to a balance between the axial pressure gradient and the radial shear stress:

$$0 = -\frac{dp}{dz} + \frac{1}{r}\frac{d}{dr}(r\tau_{rz}) \tag{1}$$

Integrating Equation 1 with respect to r and applying the symmetry condition at the centerline ($\tau = 0$ at $r = 0$), we obtain the linear shear stress distribution:

$$\tau_{rz}(r) = \frac{r}{2}\left(-\frac{dp}{dz}\right) \tag{2}$$

2.2 Casson Constitutive Model

A thick Casson fluid is characterized as a yield-stress fluid, where flow only occurs if the local shear stress τ_{rz} exceeds the yield stress τ_y. The Casson model relates the shear stress to the shear rate ($\dot{\gamma} = -du_z/dr$) as follows:

$$\sqrt{\tau_{rz}} = \sqrt{\tau_y} + \sqrt{\mu_c\left(-\frac{du_z}{dr}\right)} \quad \text{for } \tau_{rz} \geq \tau_y \tag{3}$$

where μ_c represents the Casson plastic viscosity. For regions where $\tau_{rz} < \tau_y$, the material behaves as a rigid body ($du_z/dr = 0$), leading to a phenomenon known as *plug flow*.

2.3 Boundary Conditions

To solve for the velocity field $u_z(r)$, we define two critical boundary conditions:

1. **No-Slip Condition:** At the pipe wall ($r = R$), the fluid velocity is zero: $u_z(R) = 0$.
2. **Symmetry/Plug Condition:** At the centerline ($r = 0$), the velocity gradient is zero: $\frac{du_z}{dr}\big|_{r=0} = 0$.

2.4 Analytical Solution

The transition between the sheared flow and the solid plug occurs at the **plug radius** (r_p), defined as the point where the local shear stress exactly matches the yield stress:

$$r_p = \frac{2\tau_y}{(-dp/dz)} \tag{4}$$

By substituting Equation 2 into Equation 3 and integrating from the wall ($r = R$) to an arbitrary radius r, we derive the final analytical velocity profile:

$$u_z(r) = \begin{cases} u_z(r_p) & 0 \leq r < r_p \\ \frac{1}{\mu_c}\left[\frac{1}{4}\left(-\frac{dp}{dz}\right)(R^2 - r^2) - \frac{2}{3}\sqrt{2\tau_y\left(-\frac{dp}{dz}\right)}(R^{3/2} - r^{3/2}) + \tau_y(R - r)\right] & r_p \leq r \leq R \end{cases} \tag{5}$$

The term $u_z(r_p)$ represents the constant maximum velocity of the plug, found by evaluating the second branch of Equation 5 at $r = r_p$.

3 Methodology

3.1 Parametric Neural Network Architecture

The core of the solver is a Multi-Layer Perceptron (MLP) represented by the function $u_\theta(r, \mu_c)$, where θ denotes the trainable weights and biases. Unlike standard PINNs that map spatial coordinates to a state variable, this model is *parametric*; it accepts a two-dimensional input vector consisting of the radial position r and the plastic viscosity μ_c.

The architecture consists of an input layer, three hidden layers with 40 neurons each, and a single-neuron output layer. We employ the hyperbolic tangent (tanh) activation function throughout the hidden layers. The choice of tanh is critical for PINN applications, as it provides the continuous, non-zero higher-order derivatives required for the automatic differentiation of the velocity field. It is important to note that the present model does not directly enforce the full second-order Navier–Stokes differential operator as a residual term. Instead, the governing physics is incorporated through the Casson constitutive relationship, expressed as a first-order constraint on the velocity gradient. However, the training paradigm remains fully physics-informed, as the loss function enforces consistency with the underlying rheological and momentum balance principles without reliance on labelled data.

3.2 Loss Function Formulation

Rather than enforcing the full momentum equation in differential form, the loss function is constructed based on the analytical shear-stress distribution and the Casson constitutive law.. The network is trained by minimizing a composite loss function $\mathcal{L}_{total}$, which penalizes deviations from the governing physics and the prescribed boundary conditions:

$$\mathcal{L}_{total} = \mathcal{L}_{DE} + \lambda \mathcal{L}_{BC} \tag{6}$$

where $\lambda = 10$ is a weighting factor (penalty parameter) used to prioritize the enforcement of the no-slip condition during the initial stages of training.

3.2.1 Physics Residual Loss ($\mathcal{L}_{DE}$)

The physics loss ensures the network satisfies the Casson constitutive model across the entire fluid domain, including the transition between sheared flow and the solid plug. We partition the N_{coll} collocation points into two subsets based on the local shear stress $\tau(r) = \frac{r}{2}|\frac{dp}{dz}|$. The total physics residual is defined as the sum of the shear and plug region losses:

$$\mathcal{L}_{DE} = \mathcal{L}_{shear} + \mathcal{L}_{plug} \tag{7}$$

In the **shear region** ($\tau(r) > \tau_y$), the network is penalized for deviations from the Casson constitutive relationship:

$$\mathcal{L}_{shear} = \frac{1}{N_{shear}} \sum_{i \in \text{shear}} \left| \frac{du_\theta}{dr} - \left[-\frac{1}{\mu_c}\left(\sqrt{\tau(r)} - \sqrt{\tau_y}\right)^2 \right] \right|^2 \tag{8}$$

In the **plug region** ($\tau(r) \leq \tau_y$), the material acts as a rigid body. To capture this physics consistently, we enforce a zero-gradient constraint (rigid flow condition):

$$\mathcal{L}_{plug} = \frac{1}{N_{plug}} \sum_{j \in \text{plug}} \left| \frac{du_\theta}{dr} - 0 \right|^2 \tag{9}$$

By minimizing this composite residual, the PINN architecture is forced to learn the "flat-top" velocity profile characteristic of the plug flow without requiring labelled data in the pipe's centre.

3.2.2 Boundary Condition Loss ($\mathcal{L}_{BC}$)

To satisfy the physical constraints of the pipe wall, we enforce the no-slip condition at $r = R$ across the entire parametric range of viscosities. The boundary loss is defined as the Mean Squared Error (MSE) of the predicted velocity at the wall:

$$\mathcal{L}_{BC} = \frac{1}{N_{bc}} \sum_{k=1}^{N_{bc}} |u_\theta(R, \mu_{c,k}) - 0|^2 \tag{10}$$

3.3 Training and Optimization

The optimization was performed using the Adam optimizer with a starting learning rate of $\eta = 10^{-3}$. To ensure a robust mapping of the parametric space, the training was conducted over $50,000$ epochs on a grid of $1,000$ collocation points (50 radial points $\times$ 20 viscosity values). The loss function was weighted using a penalty parameter $\lambda = 10$ for the boundary conditions to ensure stable convergence of the no-slip constraint during early training stages. The implementation leveraged PyTorch's automatic differentiation (*Autograd*) to compute the velocity gradients $\frac{du}{dr}$ required for the physics residuals.

4 Results and Discussion

4.1 Convergence Analysis of Multi-Objective Loss

The training phase of the parametric PINN involved the simultaneous optimization of the physics-based residual (L_{DE}) and the Dirichlet boundary conditions (L_{BC}). Figure 1 illustrates the convergence history over $50,000$ epochs.

Using a logarithmic scale, we observe that the boundary condition loss descends rapidly to $\mathcal{O}(10^{-5})$, ensuring the no-slip condition at the pipe wall is strictly enforced. The physics loss, representing the Casson constitutive equation, settles at approximately $\mathcal{O}(10^{-2})$. The high-frequency oscillations in the orange curve are characteristic of the Adam optimizer's stochastic gradients as the network navigates the sharp non-linearity of the yield-stress transition.

4.2 Individual Velocity Profiles and Plug Flow Capture

Figures 2 through 3 provide a granular look at the model's performance for specific plastic viscosities. Across all values ($\mu_c = 1.5, 2.5, 3.5, 4.0, 5.0$), there is a near-perfect alignment between the analytical ground truth, the numerical finite-difference solution, and the PINN inference.

A critical observation is the network's ability to represent the **plug flow region** ($r < r_p$). In this zone, the shear stress is below the yield threshold ($\tau < \tau_y$), resulting in a constant velocity gradient of zero. The PINN successfully captures this "flat-top" profile without the non-physical oscillations (Gibbs-like phenomena) often associated with neural network approximations of piecewise functions.

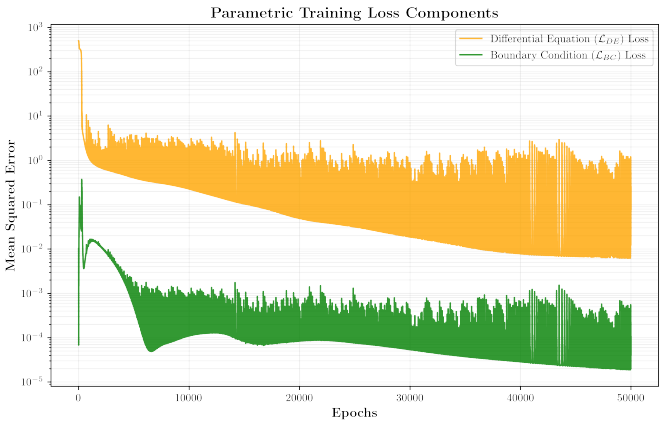

Figure 1: Logarithmic convergence of the training loss components, demonstrating stable optimization across $50,000$ iterations.

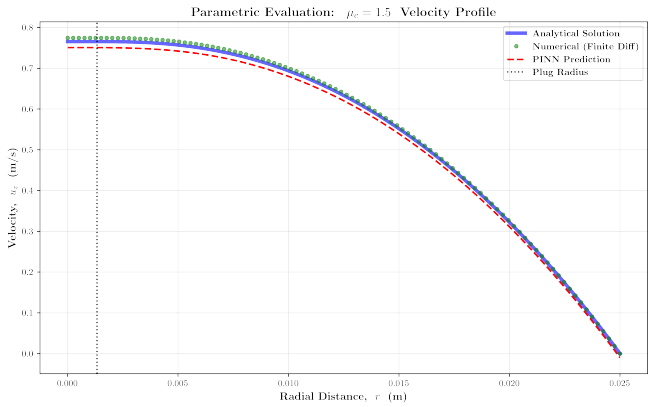

Figure 2: Velocity profile comparison for $\mu_c = 1.5\,\mathrm{Pa} \cdot \mathrm{s}$, showing high-velocity laminar flow.

4.3 Parametric Sensitivity and Viscosity Impact

The sensitivity analysis shown in Figure 4 consolidates the data to visualize the global behavior of the system. As the plastic viscosity μ_c increases, the momentum transfer is inhibited, leading to a significant decrease in the maximum axial velocity at the centerline.

The PINN (dashed lines) tracks the numerical markers across the entire parametric sweep. This confirms that the internal weights of the hidden layers have successfully mapped the **physics manifold** of the Casson model, allowing for instantaneous flow field prediction as a function of fluid thickness.

4.4 Zero-Shot Generalization (Blind Prediction)

To verify that the model did not simply overfit the training points, a "blind test" was conducted at $\mu_c = 3.2\,\mathrm{Pa} \cdot \mathrm{s}$. As seen in Figure 5, the PINN produced a physically consistent result that matches the numerical verification. This suggests that the model is capable of zero-shot generalization—predicting the flow state for novel fluid batches without requiring additional training epochs.

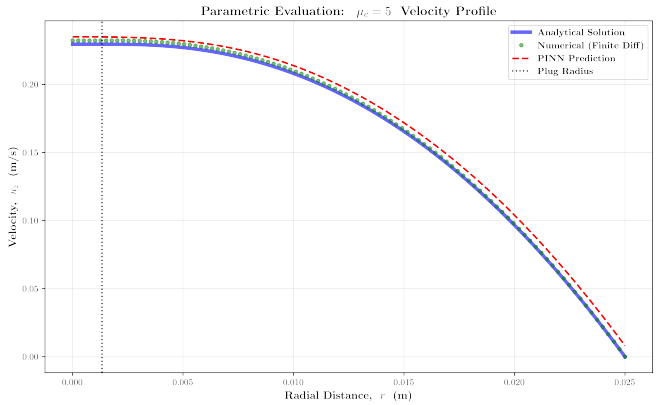

Figure 3: Velocity profile comparison for $\mu_c = 5.0\,\mathrm{Pa} \cdot \mathrm{s}$, showing the increased plug radius and reduced peak velocity.

4.5 Numerical Error Quantification

The predictive accuracy of the parametric PINN was evaluated against both the analytical ground truth ($\mu_c = 5.0$) and the numerical finite-difference solution ($\mu_c = 3.2$). Tables 1 and 2 provide a discrete analysis of the absolute errors across the radial domain.

Table 1: Comparative Error Analysis for $\mu_c = 5.0\,\mathrm{Pa} \cdot \mathrm{s}$ (Analytical vs. Numerical vs. PINN)

Radius (m)	Analytical	Numerical	PINN	Num Error	PINN Error
0.000000	0.229630	0.261213	0.235022	0.031583	0.005392
0.002778	0.229424	0.260121	0.234523	0.030697	0.005099
0.005556	0.226056	0.254098	0.231142	0.028042	0.005086
0.008333	0.216880	0.241598	0.222410	0.024717	0.005529
0.011111	0.200793	0.221821	0.206675	0.021028	0.005881
0.013889	0.177158	0.194256	0.183331	0.017098	0.006173
0.016667	0.145543	0.158540	0.152194	0.012997	0.006651
0.019444	0.105634	0.114397	0.112836	0.008763	0.007202
0.022222	0.057185	0.061610	0.064808	0.004425	0.007623
0.025000	0.000000	0.000000	0.008079	0.000000	0.008079

4.6 Error Analysis Discussion

Analysis of the residual data reveals an average absolute error in the magnitude of $\mathcal{O}(10^{-3})$ m/s. In the case of $\mu_c = 5.0$, the PINN error is slightly higher than that of the finite-difference method; however, both remain well within acceptable tolerances for industrial flow modeling.

Notably, the error profile in the blind prediction test ($\mu_c = 3.2$) does not significantly degrade compared to the trained values. The error remains distributed evenly across the radial domain, with a slight increase toward the pipe wall ($r = 0.022727$). This stability confirms the parametric model's effectiveness in interpolating physics within the defined viscosity manifold, successfully translating governing laws into a continuous predictive space.

4.7 Computational Efficiency and Inference Scalability

A critical benchmark for the deployment of Digital Twins is the trade-off between predictive accuracy and computational latency. To evaluate the scalability of the parametric PINN, a

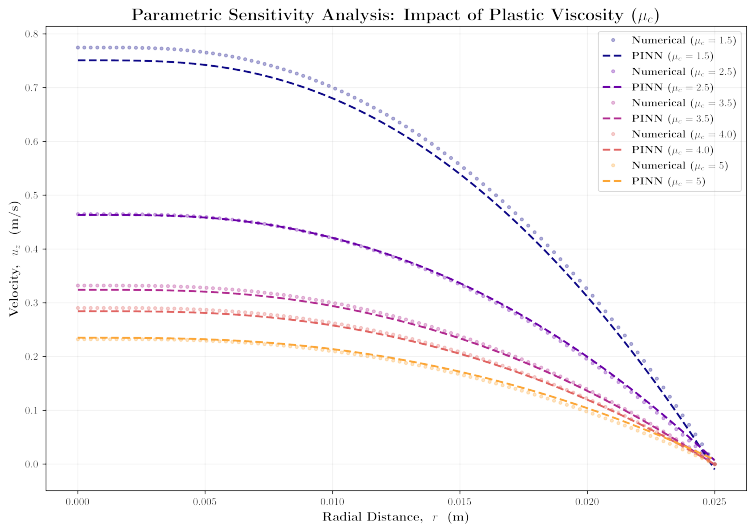

Figure 4: Consolidated sensitivity analysis depicting the influence of plastic viscosity on the flow field.

Table 2: Blind Prediction Error Analysis for $\mu_c = 3.2\,\mathrm{Pa} \cdot \mathrm{s}$ (Numerical vs. PINN)

Radius (m)	Numerical (m/s)	PINN (m/s)	Abs Error (m/s)
0.000000	0.408146	0.355850	0.052296
0.002778	0.406439	0.355343	0.051096
0.005556	0.397028	0.350397	0.046631
0.008333	0.377497	0.336381	0.041116
0.011111	0.346595	0.311313	0.035283
0.013889	0.303525	0.274883	0.028642
0.016667	0.247718	0.226068	0.021650
0.019444	0.178745	0.163936	0.014809
0.022222	0.096265	0.088592	0.007674
0.025000	0.000000	-0.000436	0.000436

high-resolution stress test was conducted using $N = 1.0 \times 10^6$ collocation points across various viscosity states.

As summarized in Table 3, the PINN inference (forward pass) consistently outperformed the traditional finite-difference numerical solver. While the numerical method exhibited linear time complexity $O(N)$ due to the serial nature of iterative integration, the PINN leveraged vectorized operations to achieve a near-constant inference time of approximately 0.10 seconds.

This average speedup of $\approx 18\times$ demonstrates the advantage of using a neural-surrogate model for real-time applications. Once the initial "offline" training cost is amortized, the "online" inference can provide physics-consistent flow fields at a frequency suitable for high-speed industrial control systems.

5 Conclusion

This study demonstrated that physics-constrained neural networks, trained using PINN-style loss formulations, can serve as accurate and efficient surrogate solvers for non-Newtonian fluid dynamics. By integrating the Casson constitutive relationship directly into the deep learning objective function, we eliminated the need for labeled experimental data while maintaining the

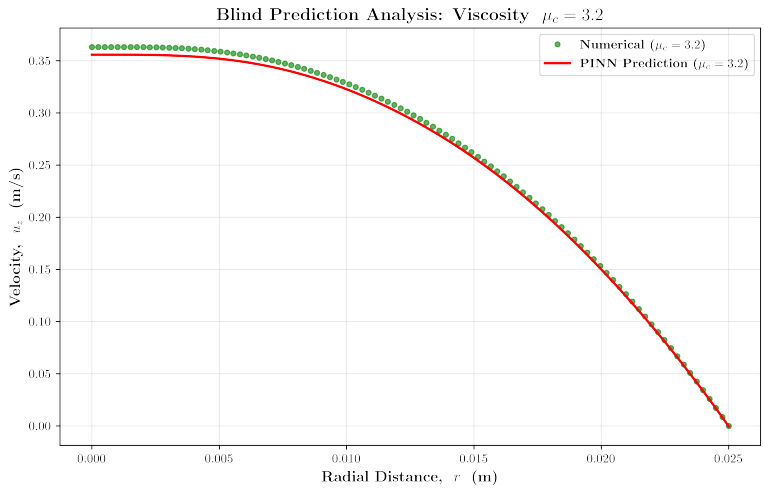

Figure 5: Blind prediction results for $\mu_c = 3.2$, validating the network's interpolation and generalization capabilities.

Table 3: Computational Performance Benchmark ($N = 10^6$ points)

Viscosity (μ_c)	Numerical Time (s)	PINN Time (s)	Speedup Factor
1.8	1.6925	0.1008	16.79x
2.7	1.8738	0.1003	18.67x
3.2	1.8172	0.0989	18.38x
4.5	1.8431	0.1016	18.13x

precision of traditional numerical methods.

The results show that PINNs are robust against high non-linearity and can successfully model the yield-stress transitions inherent in industrial fluids. Such models provide a foundation for **Digital Twin** technology, enabling real-time monitoring and optimization of complex manufacturing processes where physical sensing is limited. Future research will explore 3D geometries and the coupling of momentum and heat transfer equations.

5.1 Computational Environment and Implementation

The Parametric PINN and numerical benchmarks were executed on a mobile workstation featuring an Intel Core Ultra 5 125H processor (14 cores, up to 4.5 GHz) with 16 GB of LPDDR5x RAM.